TurnZero:
The Illusion of Adaptation When Experts Lose

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Abstract

In best-of-three Pokémon VGC tournament play, players who just lost change their lead pair far more often than winners (72% vs. 46%). This looks like adaptation, but it is not: the exploration is undirected and produces no learnable structure. We show this using permutation-invariant transformer ensembles as measurement instruments on 382K expert replays. A sequential model conditioned on prior-game outcome gains +6.6 percentage points after wins—players repeat what worked—but less than 1pp after losses. A specialist trained only on post-loss data scores below the base ensemble on that same subset (4.0% vs. 5.4%), meaning the post-loss signal actively misleads; and 87% of post-loss lead changes do not even interact with the opponent's prior leads. Stronger players change less overall, but the ∼27pp gap between post-loss and post-win change rates holds across all skill tiers.

1. Introduction

Pokémon Video Game Championships (VGC) is the official competitive doubles format. Under Open Team Sheet (OTS) rules, both players see each other's full roster (species, moves, items, abilities) before the battle. Each player then privately selects 4 of 6 Pokémon to bring and 2 of those 4 to lead, a $\binom{6}{4} \times \binom{4}{2} = 90$-way decision made before any battle begins.

In best-of-three tournament sets, the same player changes their lead pair 59% of the time between games: 72% after losing but only 46% after winning. Is this adaptation or noise? Win-stay/lose-shift is a known heuristic in simple repeated games (Nowak & Sigmund, 1993), where the action space is small enough that "shift" has a clear meaning. In a 90-way combinatorial decision, shifting could be targeted counter-play or undirected scrambling, and the distinction matters for both game theory and practical prediction. We use sequential prediction models as measurement instruments to answer the question. The input to each model is two full Open Team Sheets (8 features × 12 Pokémon); we use a permutation-invariant transformer ensemble to output a probability distribution over the 90-way lead selection. The models do not prescribe play, but their accuracy under different conditioning regimes reveals whether structured signal exists.

Three contributions:

1. We document the adaptation asymmetry behaviorally: rates, transition entropy, Elo stratification, and a directional null showing post-loss exploration is unstructured.

2. We quantify it via sequential models: conditioning on prior-game outcome yields +6.6pp after wins but ≤1pp after losses.

3. We confirm post-loss irreducibility: a specialist model trained only on post-loss data goes *below* baseline, and opponent context adds nothing.

2. Related Work

Win-stay/lose-shift is a robust heuristic in iterated games (Nowak & Sigmund, 1993). Serial dependence in competitive play, where prior outcomes bias future choices, has been studied in tennis, penalty kicks, and matching pennies (Walker & Wooders, 2001). Our contribution is measuring what "shift" looks like in a complex combinatorial decision: lots of movement, no structure.

Pokémon AI. VGC-Bench (Angliss et al., 2026) trains transformer policies on 705K battle logs, but team preview is one incidental step, never isolated or evaluated. EliteFurretAI (Simpson, 2024) initially reported 99.9% team preview accuracy, later traced to positional memorization; after random-order augmentation accuracy fell to 79%. Carli (Carli, 2025) predicts VGC leads via SVD-based matching on ∼5K logs with species-only features and no formal evaluation. Metamon (Grigsby et al., 2025) and

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PokeChamp (Karten et al., 2025) target Singles, a different format.

Set architectures. Our permutation-invariant design follows Deep Sets (Zaheer et al., 2017) and Set Transformer (Lee et al., 2019). The ensemble uses deep ensembles (Lakshminarayanan et al., 2017) for calibrated uncertainty.

3. Data and Problem Setup

Action space. Each player chooses 4 of 6 to bring ($\binom{6}{4} = 15$) and 2 of 4 to lead ($\binom{4}{2} = 6$), yielding $15 \times 6 = 90$ distinct actions.

Dataset. 212K best-of-three Regulation G tournament battles from Pokémon Showdown (Angliss et al., 2026) yield 382K directed examples. Each example pairs two full Open Team Sheets (species, item, ability, tera type, and 4 moves per Pokémon; 8 features × 12 Pokémon) with a ground-truth 90-way action. 53K linkable sets produce 136K game-to-game transitions for adaptation analysis.

Label observability. Leads are always visible. The full bring-4 is observable when all 4 Pokémon appear during the game ($\sim$80%; Tier 1). All model metrics use Tier 1 test data (n=32,328).

Split design. Teams are clustered by species composition ($\geq$4-of-6 overlap $\rightarrow$ connected components). Regime A holds out Team A variants within clusters: 247K train / 35K val / 40K test. Identical (team, opponent, action) triples are deduplicated across splits; the same matchup with different expert actions is retained as multi-modality signal.

4. The Adaptation Asymmetry

4.1. Winners Stay, Losers Explore

Across 136K game-to-game transitions, players who just won change their lead pair 46% of the time; those who just lost change 72% (χ^2 = 10,126, $p < 10^{-6}$).

The transition matrices in Figure 1 show the gap clearly. After winning, 91% of transitions stay on the diagonal (same leads), with conditional entropy $H = 0.52$ nats. After losing, diagonal concentration drops to 77% and entropy nearly doubles to $H = 1.02$ nats. Losers explore a 2× wider effective action space.

4.2. Exploration is Unstructured

If post-loss adaptation were strategic counter-play, lead changes should respond to what the opponent did. In mirror matchups (teams sharing species), a player can promote a shared species into their lead slot, moving *toward* the opponent's game-1 leads, or drop one, moving *away*. Among post-loss changers, 87% of lead switches do not change the overlap with the opponent's prior leads at all (n=43,063; 58.2% win rate). Of the 13% that do, switching toward and switching away produce identical game-2 win rates (54.8% vs. 54.9%, $n\approx$3,100 each). Losers are scrambling, not counter-adapting.

Table 1. Lead change rates by Elo quartile. Stronger players change less in both conditions, but the gap ($\sim$27pp) is stable across skill levels. $\chi^2 = 863$, $p \approx 0$.

Elo tier	After loss	After win	Gap
Q1 (low, <1119)	78.6%	53.3%	25.3pp
Q2 (1119–1197)	75.0%	46.9%	28.0pp
Q3 (1197–1303)	69.9%	42.6%	27.3pp
Q4 (high, >1303)	64.3%	37.3%	27.1pp

4.3. Skill Does Not Explain It

Table 1 stratifies by player Elo. Stronger players are more consistent in both conditions (Q4 players change leads 27pp less often than Q1), but the *gap* between post-loss and post-win rates is stable at $\sim$27pp across all tiers. This is not a low-skill artifact.

4.4. Null Controls

Speed control mode (Trick Room vs. Tailwind) does not explain the asymmetry: the entropy difference between these team types is $\Delta H = 0.016$ nats, effectively zero. Per-team entropy terciles show similar change rates across structurally rigid and flexible teams. The asymmetry is outcome-driven, not team-driven.

5. Measuring the Asymmetry with Prediction Models

5.1. Base Ensemble

Each of 12 Pokémon is tokenized as the sum of 8 learned embeddings (species, item, ability, tera type, 4 moves) and processed by a 4-layer transformer encoder (d=128, H=4, d_{ff}=512). Tokens are canonically sorted and aggregated via mean pooling with no positional encoding, avoiding the leakage that inflated prior work from 99.9% to 79% (Simpson, 2024). A 5-member deep ensemble (Lakshminarayanan et al., 2017) averages probabilities across independently trained members (5.8M total parameters, $\sim$5 min per member on one RTX 4080 Super). Each member is trained for 50 epochs with AdamW (lr $= 3 \times 10^{-4}$, batch size 512, cosine schedule with 5-epoch warmup); hyperparameters were selected on the validation split.

Performance. The ensemble achieves 6.4% top-1 and 15.5% top-3 on the 90-way task (vs. 1.1% random). This

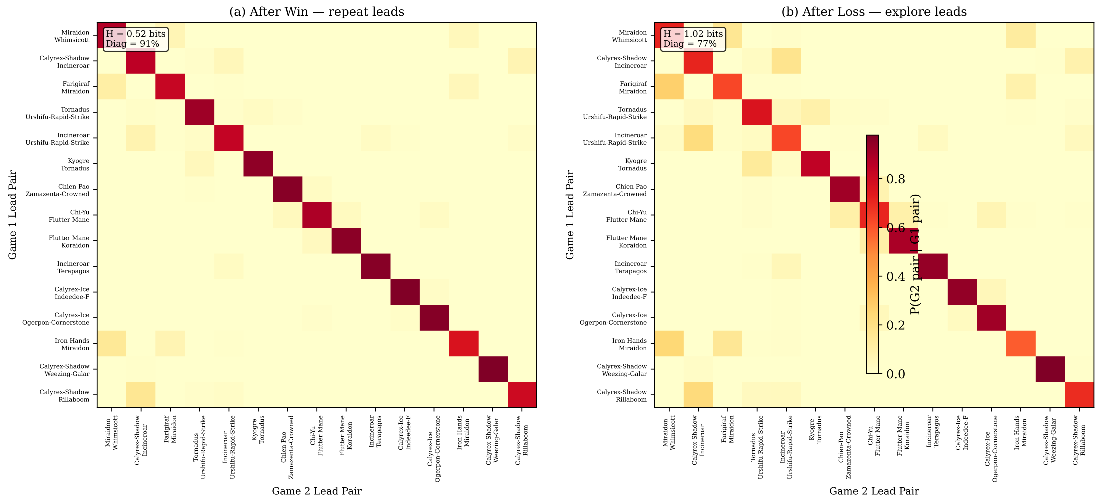

Figure 1. Lead-pair transition matrices for the 15 most common pairs, conditioned on game-1 outcome. **Left:** After winning—91% of mass on diagonal ($H = 0.52$ nats). **Right:** After losing—only 77% diagonal, entropy doubles ($H = 1.02$ nats). Winners repeat; losers scatter.

is low in absolute terms, but the task has 90 classes and experts often disagree with each other. The question is not whether the model is "good" but what its accuracy pattern reveals about the structure of play. Adding 34% more parameters changed accuracy by only ±0.3pp—well within noise—which means the model is not too small for the task; there is just no more signal to find. Validation and test accuracy differ by <0.5pp across all models, and the ensemble is well-calibrated (adaptive ECE $= 0.012$), so overfitting is not a concern.

5.2. Convention, Not Strategy

An opponent ablation experiment reveals where the base model's signal comes from. Removing the opponent's team at inference (all features masked to UNK) costs only -1.3pp top-1 (6.4% → 5.1%); removing the player's own team collapses accuracy to 1.1%, indistinguishable from random. The model retains ∼80% of its performance from self-team features alone. It learns *what this team usually does*, not *how it responds to a specific opponent*. The mirror/non-mirror gap tells the same story: accuracy on mirror matchups (common archetypes with many training examples) is 6.8%, versus 3.8% on novel pairings. Convention is team-specific and requires repeated observation; without enough training examples the model has nothing to latch onto.

Per-team heterogeneity. The 6.4% aggregate masks wide variation across 153 team archetypes (≥20 Tier 1 examples each; Figure 2). At one extreme, Commander teams (e.g., Dondozo + Tatsugiri) reach 50% top-1. The Commander ability mechanically requires both Pokémon to lead together, which concentrates the action distribution heavily ($H = 3.10$ nats). At the other extreme, flexible "goodstuffs" teams with many viable configurations sit at 0% top-1 ($H = 4.41$ nats). Between these poles, 153 archetypes trace a clear trend: the entropy–accuracy correlation is $r = -0.56$ across all archetypes and $r = -0.77$ for well-sampled archetypes ($n \geq 50$). Predictability tracks how concentrated a team's action distribution is, not strategic sophistication. Speed control mode (Trick Room vs. Tailwind) contributes nothing ($\Delta H = 0.016$ nats); and the top-17 predictions cover 50% of expert actions across the full test set, which narrows the effective decision space from 90 to 17 for the most conventional half of play.

If the base signal is convention, the adaptation asymmetry should be convention-shaped too. It is.

5.3. Sequential Model Reveals the Asymmetry

If the base model captures convention, adding game history should help only where history is informative—after wins, where players repeat. A sequential variant tests this by prepending a single context token encoding game number (1, 2, or 3), prior result (win, loss, or none), and the prior lead pair index (90-way, or a sentinel for game 1). This adds only 2,816 parameters (<0.3% of the base model), and the sentinel for game 1 ensures the context introduces no phantom signal. Table 2 shows the result.

The sequential model recovers +6.6pp after wins (6.4% → 13.0%) but only +0.8pp after losses (5.4% → 6.2%). Game-1 accuracy is unchanged (6.6% base vs. 6.4% sequential), so the context mechanism introduces no leakage. Even with history, the model reaches only 9.4% top-1 on games 2–3. The ceiling is lower after wins, but post-loss remains a wall.

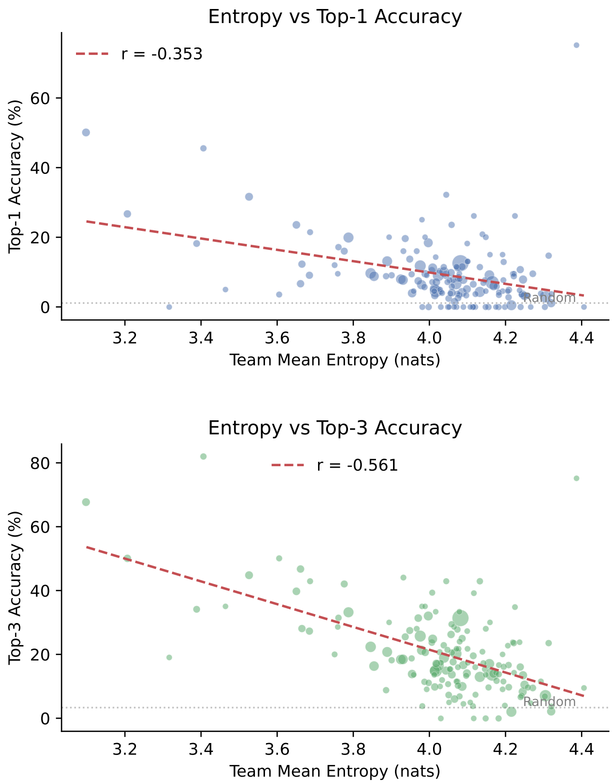

Figure 2. Per-archetype accuracy vs. mean action entropy for 153 team archetypes (≥20 Tier 1 examples). Commander teams cluster in the low-entropy / high-accuracy corner; flexible "goodstuffs" teams spread across the high-entropy floor. Top-3 correlation: $r = -0.56$ (all), $r = -0.77$ ($n \geq 50$).

Table 2. Top-1 accuracy (%) on 90-way prediction, stratified by prior outcome. Sequential gains come entirely from post-win. Opponent context adds nothing. The post-loss specialist performs *worse* than the base model on its own target data (4.0% vs. 5.4%).

Model	Params	After win	After loss	G2–3
Base ensemble	5.8M	6.4	5.4	5.9
Sequential	+2.8K	**13.0**	6.2	**9.4**
+ Opp. context	+4.1K	12.6	6.0	9.2
Post-win spec.	+2.8K	10.1	3.9	—
Post-loss spec.	+2.8K	7.0	4.0	—

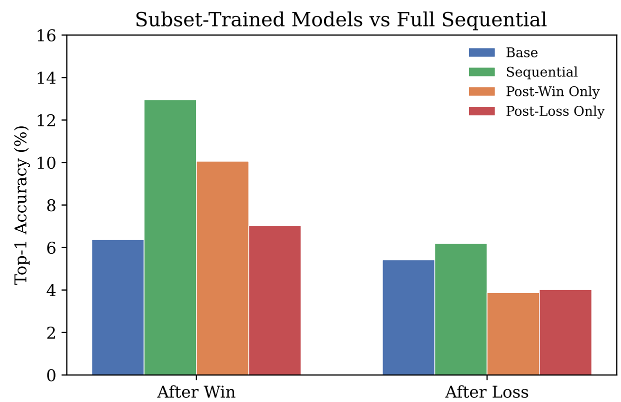

Figure 3. Post-win specialist captures real signal (10.1% vs. base 6.4% after wins). Post-loss specialist falls *below* base (4.0% vs. 5.4% after losses)—there is no learnable structure in post-loss adaptation.

5.4. Post-Loss is Irreducible

Maybe post-loss behavior is learnable but the sequential model just lacks the right information. Three experiments rule this out.

Opponent context is null. Adding the opponent's revealed game-1 species as additional context tokens (+4.1K parameters) produces no post-loss improvement (6.0% vs. 6.2% for the sequential model without opponent context) and slightly *hurts* post-win accuracy (12.6% vs. 13.0%). Whatever signal the opponent's revealed species might carry is already captured by the base sequential model's own-team context.

Post-loss specialist goes negative. A model trained exclusively on post-loss examples achieves only 4.0% on its own target data, worse than the base model's 5.4% on the same subset (Figure 3). This is not a sample-size problem: the post-loss training set contains ∼18K examples, plenty for a 90-class task. There is no pattern to learn. The post-win specialist, by contrast, reaches 10.1% after wins—post-win repetition is real and learnable.

Directional null. As shown in Section 4.2, the direction of post-loss lead changes carries no predictive signal for game-2 outcome.

6. Discussion

Why the asymmetry. Winners have confirmed information ("this worked") and rationally exploit it by repeating. The win-stay signal is strong because it requires no inference: if the lead pair won, repeat it. Losers face a harder problem. Something failed, but the team preview decision is entangled with dozens of in-game choices (switches, move selection, tera timing), and the player cannot attribute the loss to leads specifically. So they change—72% do—but without a clear causal signal, the change is undirected. The 90-way action space is large enough that undirected shifts wash out in aggregate: each loser scrambles in their own direction. This is a credit-assignment problem. In a game with hundreds of micro-decisions per battle, leads are the most visible lever to pull but rarely the actual cause of a loss.

Convention, not strategy. The base model learns what teams typically do (opponent ablation: -1.3pp), and the sequential model learns who repeats (post-win: +6.6pp). Neither learns opponent-conditional adaptation (opponent context: null). A 6.4% top-1 is what accuracy looks like when experts disagree about the right play, and the predictability that does exist comes from mechanical constraints (Commander teams at 50%) rather than strategic reasoning.

Limitations. Bootstrap confidence intervals are wide (top-1: [2.6%, 6.6%]), reflecting real heterogeneity across team archetypes. Results are specific to Regulation G; whether the adaptation asymmetry generalizes to other formats or competitive domains is untested. Data is exclusively best-of-three tournament play; ladder games were deliberately excluded to match deployment conditions but limit scale. The adaptation effectiveness question (does changing leads *help*?) is confounded by selection: players who stay after a loss may have lost due to misplay rather than bad leads, so we cannot make causal claims without randomization. Some players in the dataset lack Elo ratings.

What next. Other VGC regulations, with different restricted Pokémon pools, may shift the balance between convention and flexibility—testing replication there is the obvious next step. Beyond Pokémon, any competitive domain with combinatorial team selection and best-of-N structure (MOBA drafting, fighting game counterpicks, sports lineup decisions) could show analogous patterns. A lab or simulation setting where lead choice can be randomized would let us ask the causal question our observational data cannot: does post-loss adaptation actually help?

Contributions. Solo project. Code:
https://github.com/leba01/turnzero
Demo: https://turnzero.vercel.app

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